

Lab 3.3.11 Report: Exploring Processes, Threads, Handles, and Windows Registry

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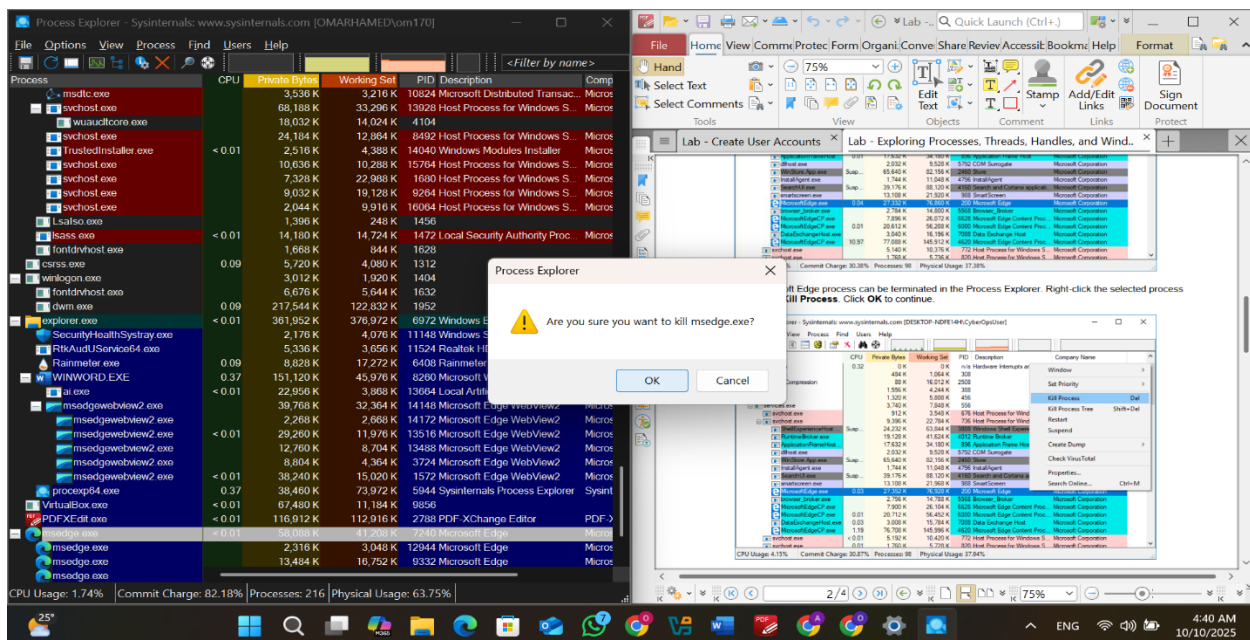
Part 1: Exploring Processes

Question 1: What happened to the web browser window when the process is killed?

Answer:

When the browser process was killed, the web browser window closed immediately.

This happened because the process is the running instance of the application, and when terminated, all windows and resources associated with it are closed.



Question 2: What happened during the ping process?

Answer:

During the ping process, a new child process (PING.EXE) appeared under cmd.exe.

The CPU usage increased slightly, and network activity was visible. The process

showed active threads handling the ICMP packets being sent and received.

Question 3: What happened to the child process conhost.exe?

Answer:

When the parent process (cmd.exe) was killed, the child process (conhost.exe) was also terminated automatically. This is because conhost.exe depends on cmd.exe, and when the parent dies, Windows terminates all dependent child processes.

Part 2: Exploring Threads and Handles

Question 4: What type of information is available in the Properties window?

Answer:

The Properties window shows detailed information about threads including:

- Thread ID (TID)
- CPU usage and cycles consumed by each thread
- Start time of each thread
- Thread state (Running, Waiting, etc.)
- Starting address of the thread
- Module information

This helps in understanding how the process distributes work across multiple threads.

Question 5: Examine the handles. What are the handles pointing to?

Answer:

The handles are pointing to various system resources including:

- File handles (files opened by the process)
- Registry keys

- Synchronization objects (Events, Mutexes, Semaphores)
- Devices and drivers
- Memory sections
- Threads

These handles represent references to operating system resources that the process is using.

Part 3: Exploring Windows Registry

Question 6: What is value for this registry key in the Data column?

Answer: (1)

After changing the value from 1 to 0, the Data column shows: 0x00000000 (0)

This indicates that the EULA has not been accepted.

Question 7: When you open the Process Explorer, what did you see?

Answer:

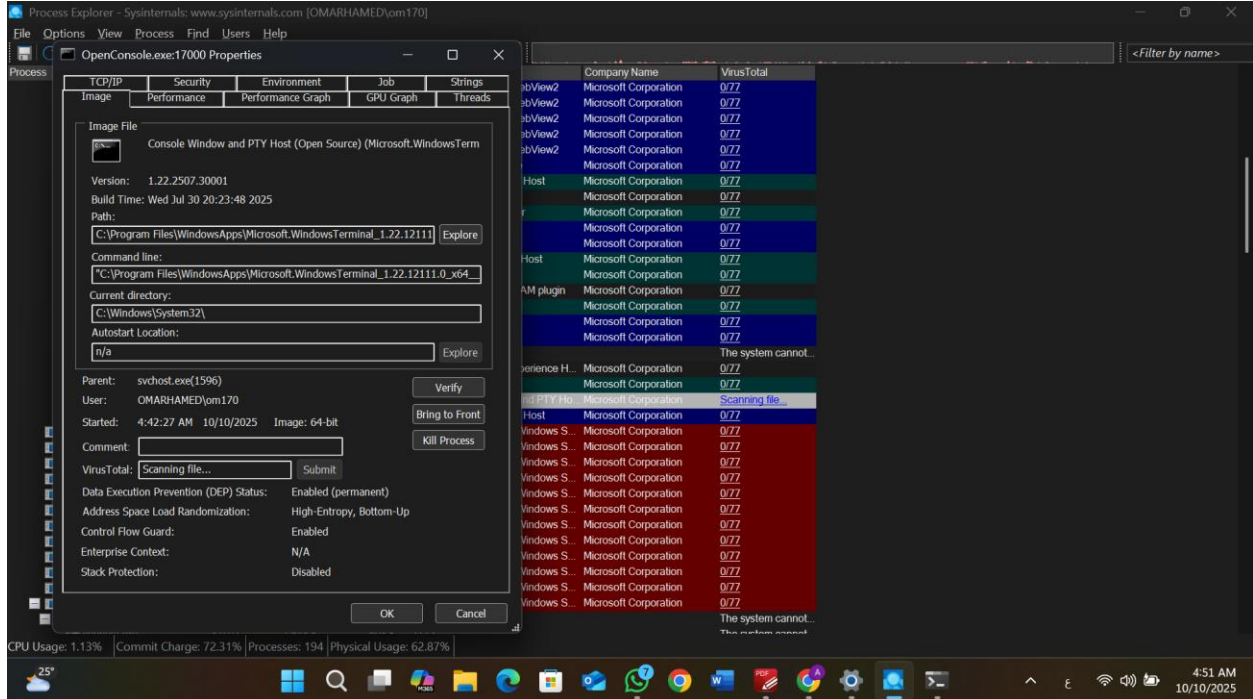
When I opened Process Explorer after changing the registry value to 0, the program appeared to be deleted or corrupted. A dialog box appeared asking if I wanted to remove the shortcut because the program seemed to be missing or inaccessible. I went back to the Registry Editor and manually changed the value back to 1. After restoring the registry value, Process Explorer launched successfully and worked normally. This shows that the registry key is critical for the program's proper operation, and incorrect values can prevent the application from running correctly.

Conclusion:

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This lab was an eye-opening experience that taught me practical skills in Windows system management. I discovered how the Windows Registry controls application behavior, specifically how I can manage software licenses by modifying registry keys without uninstalling programs. This gives me much more control over my system configuration.

Learning to use Process Explorer was particularly valuable. I can now see all processes running on my computer and identify which ones are legitimate. The VirusTotal integration is incredibly useful—knowing that a 0/77 score means the process is clean (zero threats detected by 77 antivirus programs) gives me confidence in my system's security.



Most importantly, I now have the tools and knowledge to detect malicious processes on my system before they cause damage. This lab transformed my

understanding of how Windows manages processes and gave me practical cybersecurity skills I can use immediately.